

Muzammil Nawaz Khan

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Professional Summary

Versatile AI Engineer with strong expertise in **Generative AI**, **Voice Agents**, and **Computer Vision**. Proven experience in building real-time conversational agents using OpenAI and LiveKit, as well as optimizing deep learning models for biomedical image segmentation (Winner, COMPPEC 2025). Skilled in Python, PyTorch, and deploying automated workflows to solve complex enterprise challenges.

Technical Skills

AI & Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, OpenAI API, LLMs; **Voice & Agents:** LiveKit, Silero VAD, ElevenLabs, n8n (MCP Server), SIP Trunking, STT/TTS; **Computer Vision:** OpenCV, U-Net, SegFormer, YOLO, Object Detection; **Languages & Tools:** Python, C++, SQL, Git, Docker, NumPy, Pandas.

Professional Experience

CareCloud | AI Engineer | July 2025 – Present

- Architected a real-time Voice AI agent using LiveKit and ElevenLabs, integrating Silero VAD for low-latency interruptions, reducing front-desk handling time by 40%.
- Designed an n8n-based orchestration layer acting as an MCP server to manage state and logic-based routing, decreasing inference costs by 35% through efficient prompt caching.
- Engineered automated Python pipelines for Hikvision NVR systems, implementing logic filters to clean noise and improving HR attendance tracking accuracy by 25%.
 - **Agent Orchestration:** Implemented an n8n workflow acting as a Model Context Protocol (MCP) server to manage conversation state, backend prompts, and call routing logic.
 - **Telephony Integration:** Configured **SIP Trunking** to handle inbound calls with logic-based transfer functionality to human agents.
 - **Data Pipeline Automation:** Engineered Python scripts to interface with Hikvision NVR systems, fetching raw logs for facial recognition and optimizing workforce analytics.

Risetech Islamabad, Pakistan *AI Research Intern* July 2024 – Aug 2024 – **Neurofusion Project (FYP):** Spearheaded the development of a 3D brain tumor segmentation system using the BraTS20 dataset.

- **Model Optimization:** Benchmarked **3D U-Net** against **SegFormer3D**, selecting the latter to achieve **85%+ accuracy**. – **Awards:** Won **1st Place at COMPPEC 2025** and received the Departmental Silver Medal for technical complexity.

Projects

Retinal Image Segmentation | Python, OpenCV

- Developed multi-level thresholding and morphological segmentation methods to automate vessel visibility enhancement for medical analysis.

Optical Braille Recognition | Python, OpenCV

- Built a dot-pattern recognition system using noise filtering and contour analysis to translate Braille characters with high precision.

Skin Image Segmentation | Python, OpenCV

- Designed a skin lesion segmentation pipeline using Connected Component Labeling (CCL) and post-processing for clinical diagnostic support.

Education

National University of Sciences and Technology (NUST) | Bachelor of Computer Engineering | Nov. 2021 – July 2025

- Islamabad, Pakistan | CGPA: 3.25
- Relevant Coursework: Digital Image Processing, AI and Decision Support System, Computer Vision

Ghazali Premier College | F.Sc. Pre-Engineering | June 2018 – Aug 2020

- Lahore, Pakistan

Professional Certifications

- **IBM** AI Engineering Professional Certificate (Deep Learning, PyTorch, Keras)
- **IBM** Generative AI Engineering Professional Certificate (LLMs, Transformers)
- **IBM** RAG & Agentic AI Professional Certificate (LangChain, LangGraph)
- LLMOps (Large Language Model Operations) Specialization – **Duke University**
- Informed Clinical Decision Making using Deep Learning – **University of Glasgow**
- Modern Data Strategy for Enterprise GenAI – **Fractal**
- Complete Modern C++ (C++11/14/17) – **Packt**